



HELWAN NATIONAL UNIVERSITY

Faculty of Engineering · Department of Mechatronics & Robotics Engineering

QUALITY CONTROL

*DC Motor Manufacturing Process Analysis
Using Statistical Process Control (SPC)*

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AGENDA

01

Introduction & Data

30 batches · $n=5$ · Winding Resistance

04

Root Cause (Fishbone)

6M framework: all possible causes

07

Revised Charts

Verification after improvements

02

X-bar & R Charts

Process stability: mean & variation

05

Pareto Analysis

Ranking causes by frequency

08

Normality Test & Cpk

Capability vs specification limits

03

P-Chart Analysis

Proportion of defective units

06

Process Improvement

Corrective actions taken

09

Comparison & Conclusion

Before vs After + final takeaways

INTRODUCTION · Quality Control in DC Motor Manufacturing

What is Winding Resistance (WR)?

A critical quality characteristic measured in Ohms (Ω). Abnormal values indicate poor winding, overheating, material inconsistency, or internal electrical defects.

SPC Tools Used

X-bar & R Charts, P-Chart, Fishbone Diagram, Pareto Chart, Revised Charts, Normality Test, Process Capability (Cpk), and Run Chart.

Project Objective

Detect process instability, identify assignable causes of variation, and propose corrective actions to improve manufacturing quality.

30

Production
Batches Analyzed

5

Motors Inspected
per Batch (n)

150

Total Units
Inspected

15

Total Defectives
Found

METHODOLOGY · SPC Analysis Workflow

STEP 1

Data Overview

Understand dataset & spec limits

STEP 4

Fishbone Diagram

Investigate all possible causes

STEP 7

Revised Control Charts

Verify stability after fixes

STEP 2

X-bar & R Chart

Detect out-of-control points

STEP 5

Pareto Chart

Focus on the vital few

STEP 8

Process Capability Cpk

Prove process meets spec limits

STEP 3

P-Chart

Quantify defect proportion

STEP 6

Process Improvement

Apply corrective actions

STEP 9

Run Chart

Visualize trends over time

X-bar & R CHARTS · Control Limit Calculations

R CHART (Range Chart)

Subgroup size $n = 5$

Constants (Appendix VI) $D_3 = 0 \mid D_4 = 2.114$

ΣR (Sum of Ranges) 4.62Ω

$R = \Sigma R / 30$ 0.154Ω

$UCL_R = D_4 \times R$ **0.3256Ω**

$CL_R = R$ **0.154Ω**

$LCL_R = D_3 \times R$ **0**

$\bar{X}$ CHART (Mean Chart)

Subgroup size $n = 5$

Constant (Appendix VI) $A_2 = 0.577$

$\Sigma \bar{X}$ (Sum of Means) 135.51Ω

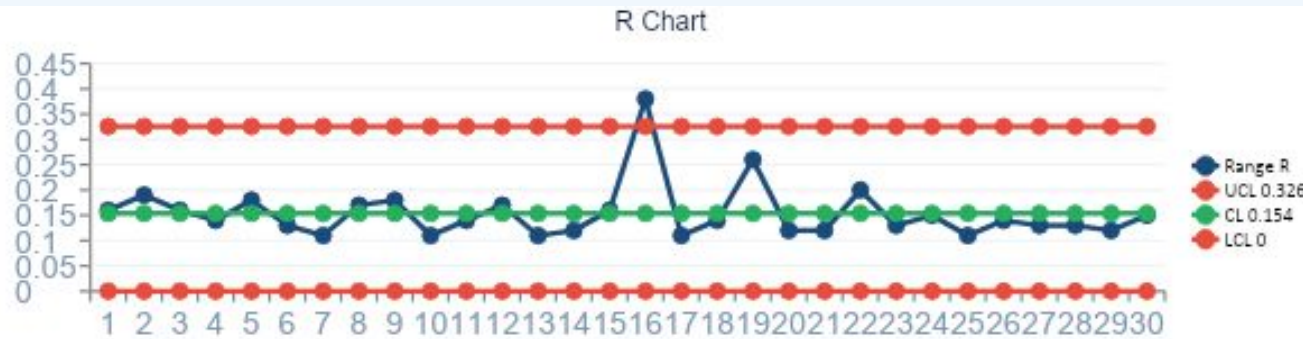
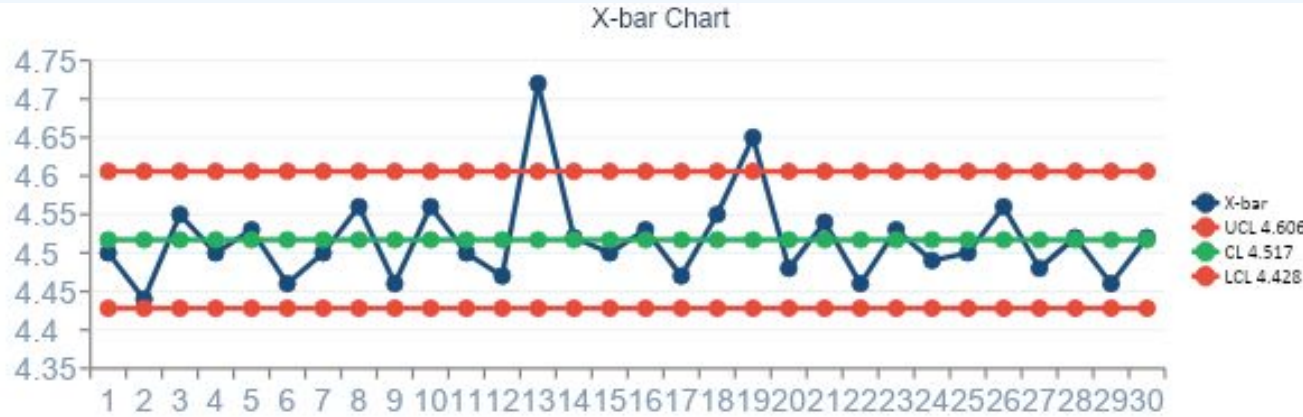
$\bar{X} = \Sigma \bar{X} / 30$ 4.517Ω

$UCL_{\bar{X}} = \bar{X} + A_2 \times R$ **4.6059Ω**

$CL_{\bar{X}} = \bar{X}$ **4.517Ω**

$LCL_{\bar{X}} = \bar{X} - A_2 \times R$ **4.4281Ω**

X-bar & R CHARTS · Results & Interpretation



KEY FINDINGS

B13 - Out of Control

X-bar = 4.720 > UCL (4.606)
Process mean shifted UP

B19 - Out of Control

X-bar = 4.650 > UCL (4.606)
Process mean shifted UP

B16 - R Chart Violation

R = 0.380 > UCL (0.326)
Abnormal within-batch spread

Process Verdict

NOT in statistical control
Special causes present

P-CHART ANALYSIS • Proportion of Defective Units

CALCULATIONS

$$p = np / n$$

Σnp (Total defectives)

15

$$\Sigma n = 30 \times 5$$

150

$$\bar{p} = 15 / 150$$

0.10

$$UCL_p = \bar{p} + 3\sqrt{(\bar{p}(1-\bar{p}))/n}$$

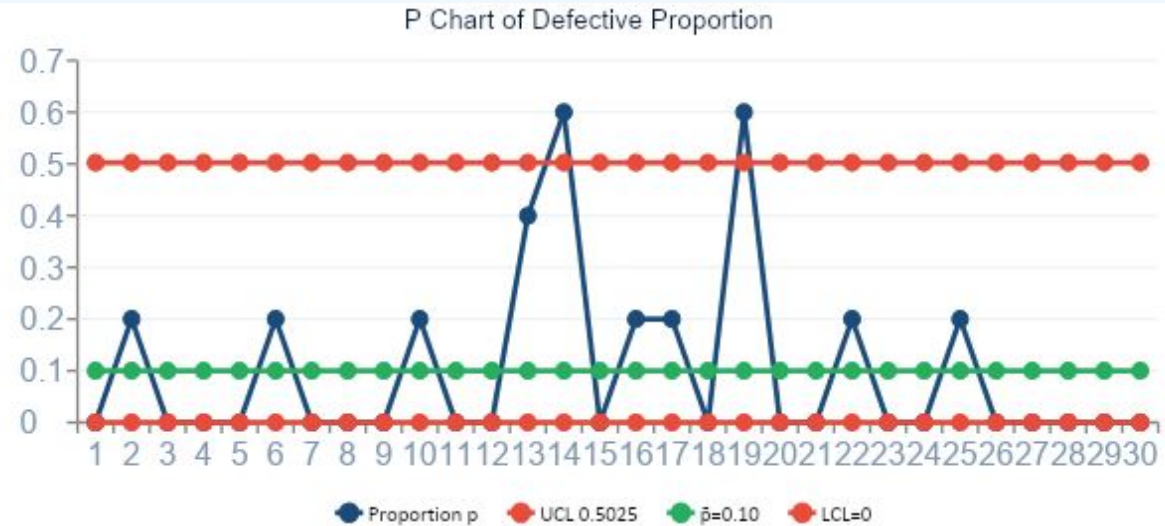
0.5025

$$CL_p = \bar{p}$$

0.10

$$LCL_p (= -0.3025 \rightarrow 0)$$

0



INTERPRETATION

Batches B14 and B19 exceeded UCL (0.5025) with $p = 0.60$ — abnormally high defect proportions.
Process is UNSTABLE and confirms X-bar/R chart findings.

ROOT CAUSE ANALYSIS · Fishbone (Ishikawa) Diagram

Analysis focused on out-of-control batches: B13, B14, B16, B19 · 6M Cause Framework

MACHINE

- Cooling system failure
- Worn forming tools
- Bearing wear (M2)
- No preventive maintenance

ENVIRONMENT

- High ambient temperature
- No temp monitoring
- Inadequate ventilation
- Poor workstation layout

MEASUREMENT

- No inline WR sensor
- Equipment uncalibrated
- Non-standard criteria
- Manual data recording

MATERIAL

- Wire resistance variation
- Magnet bond softening
- Insulation coat QC
- Supply chain delays

MAN / METHOD

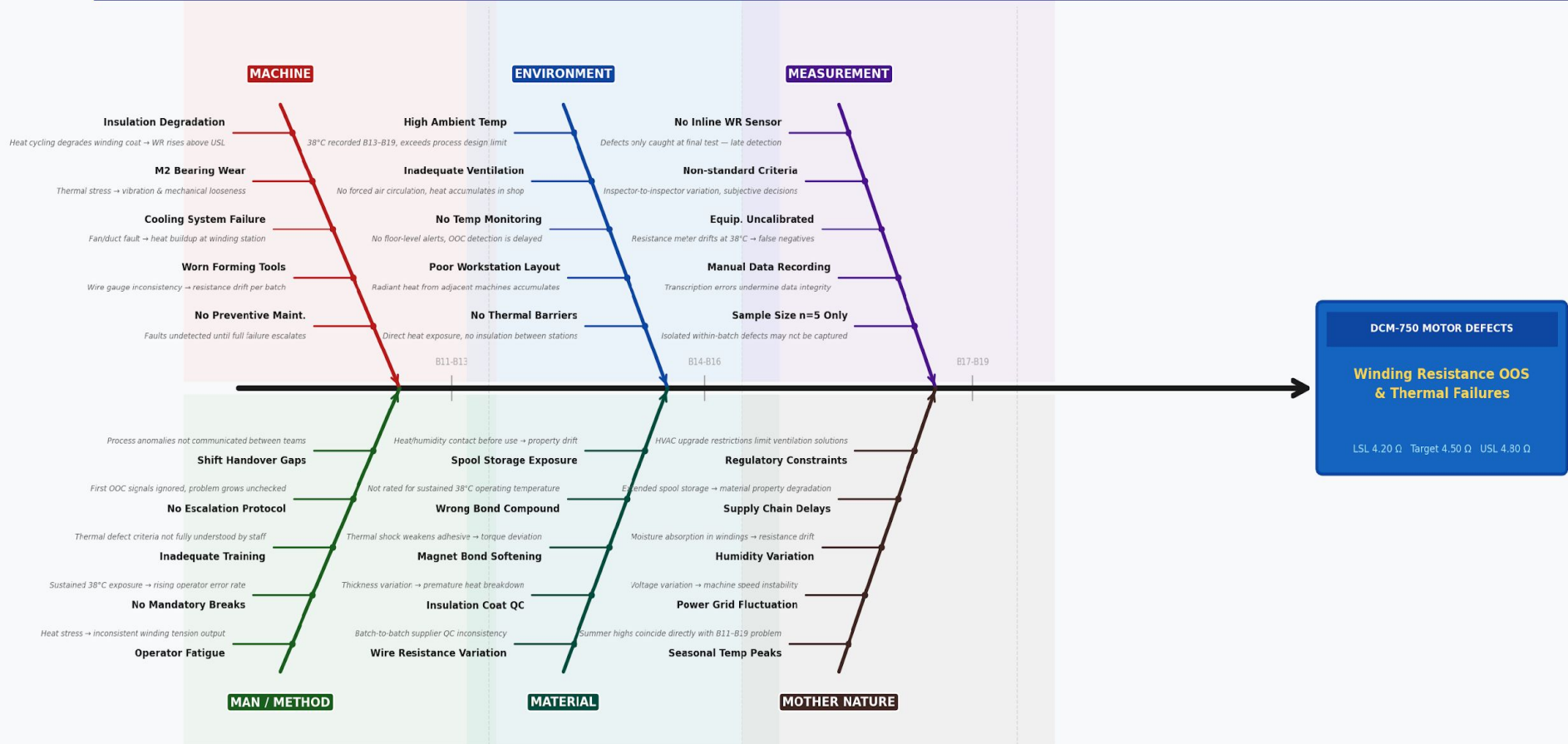
- Shift handover gaps
- Operator fatigue
- No escalation protocol
- Inadequate training

MOTHER NATURE

- Power grid fluctuation
- Seasonal temp peaks
- Humidity variation
- Regulatory constraints

DCM-750 DC Motor Quality Control — Fishbone (Ishikawa) Cause-and-Effect Diagram

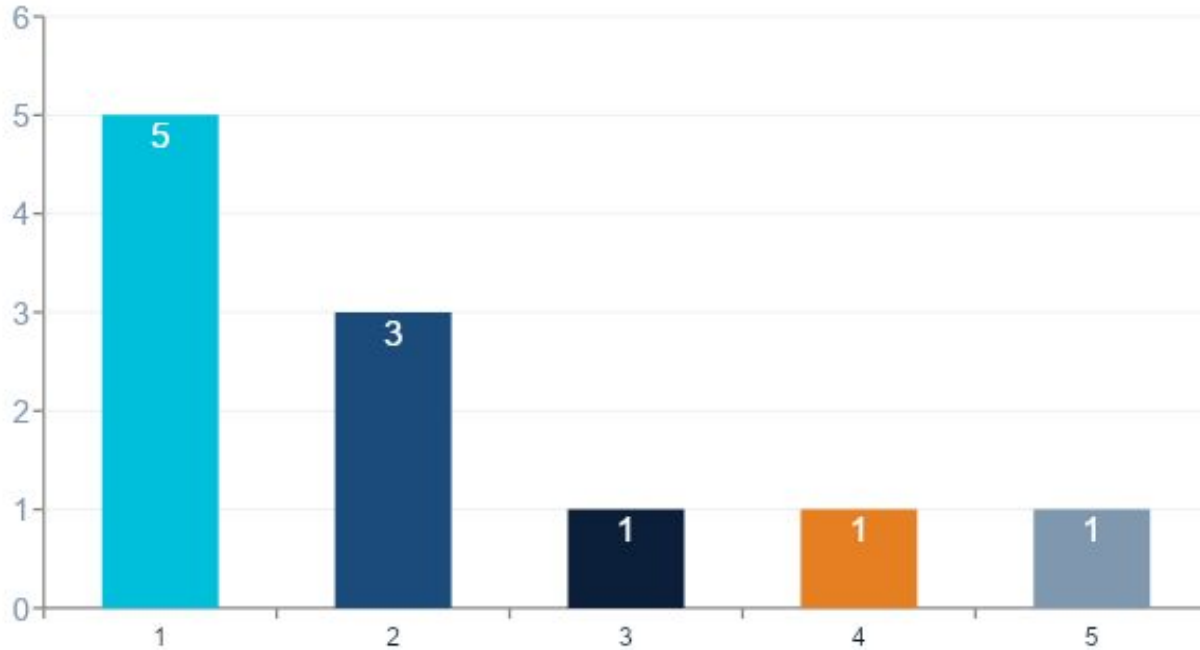
Effect: Winding Resistance Out-of-Spec (WR > USL 4.80 Ω) & Thermal Motor Failures | Problem Phase B11-B19 | 6M Framework | n=5 per batch



Branches: MACHINE · ENVIRONMENT · MEASUREMENT · MAN/METHOD · MATERIAL · MOTHER NATURE | Sub-causes confirmed from OOC analysis of B13, B14, B16 (R), B19

PARETO ANALYSIS · Identifying the Vital Few Causes

Defect Cause Frequency



PARETO TABLE

● Thermal Issues	45.5%
45.5% (cum: 45.5%)	
● Material Defect	27.3%
27.3% (cum: 72.7%)	
● Machine Fault	9.1%
9.1% (cum: 81.8%)	
● Operator/Method	9.1%
9.1% (cum: 90.9%)	
● Measurement	9.1%
9.1% (cum: 100%)	

VITAL FEW: Thermal Issues (45.5%) + Material Defects (27.3%) = **72.7%** of all defects → Focus improvement here!

PROCESS IMPROVEMENT ACTIONS · Corrective Measures

Thermal Control Improvements

PRIMARY CAUSE — 45.5% of defects

- Improved cooling systems installed
- Enhanced workshop ventilation
- Real-time temperature monitoring

Material Quality Control

SECONDARY CAUSE — 27.3% of defects

- Standardized insulation coating
- Rigorous raw material inspection
- Supplier quality verification

Equipment Maintenance

SUPPORTING ACTIONS

- Preventive maintenance schedules
- Inspection & replacement of worn parts
- Calibration of all instruments

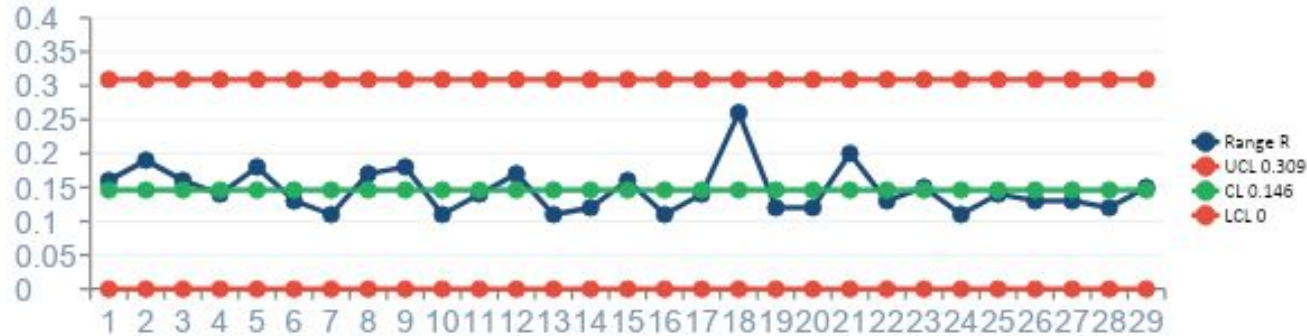
Process & Operator Training

SUPPORTING ACTIONS

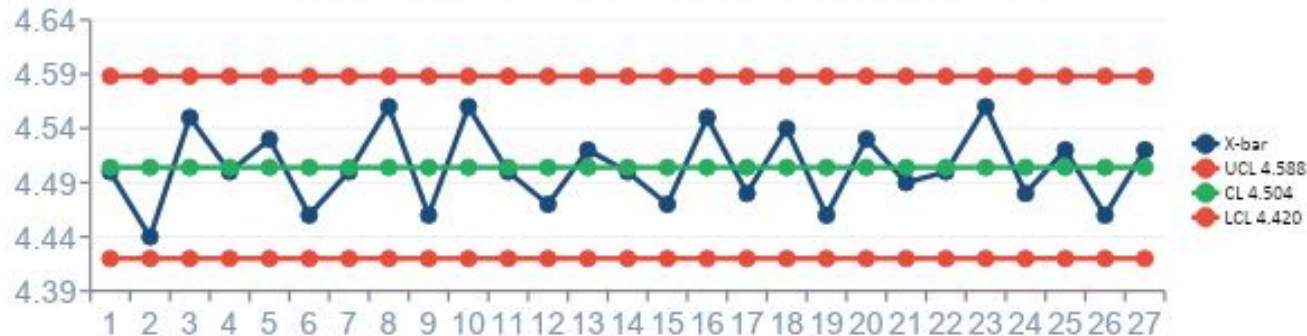
- Operator training programs
- Clear escalation procedures
- Reduced manual errors via SPC

REVISED CONTROL CHARTS · After Implementing Corrective Actions

Revised R Chart (B16 excluded — all points within limits)



Final X-bar Chart (B13, B16, B19 excluded — all points within limits)



RESULTS



R Chart

All 29 points in range
Variation STABLE



Final X-bar

All 27 points in range
Mean STABLE



Revised P-chart

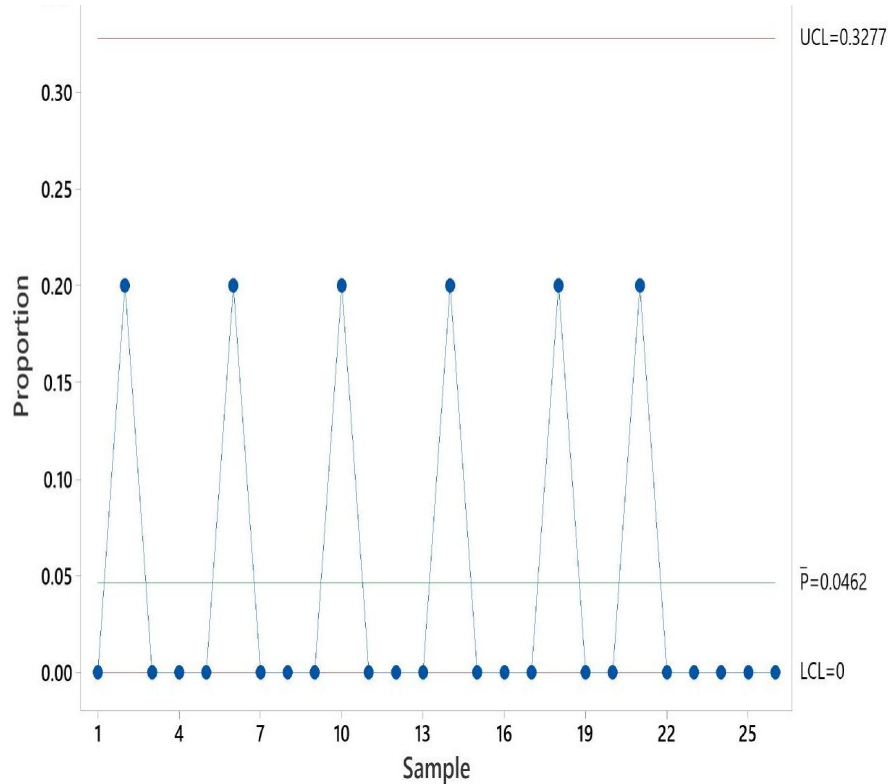
$\bar{p}$: 0.10 $\rightarrow$ 0.0462
Defect rate reduced



Process Status

FULLY in statistical
control

REVISED P-CHART · After Corrective Actions (Section 9.9.2 · Minitab)



RESULTS

Source

Section 9.9.2
Minitab Revised P-Chart

Batches

26 stable batches
(B13,B14,B16,B19 removed)

UCL

0.3277

CL ($\bar{p}$)

0.0462

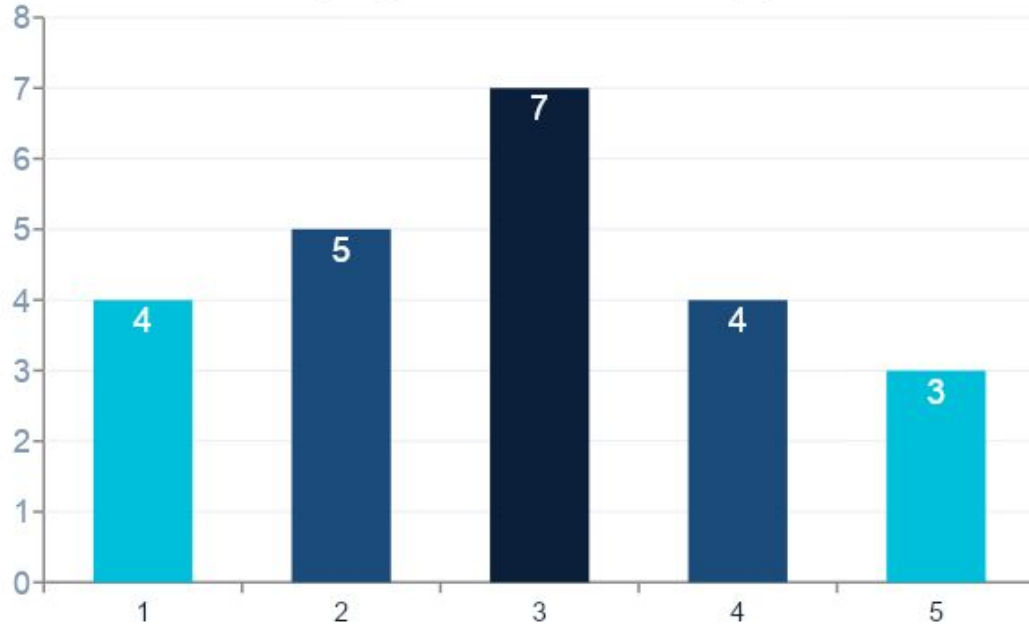
LCL

0 (negative $\rightarrow$ 0)

✓ All 26 points within limits — Defect rate improved: $\bar{p} = 0.10 \rightarrow 0.0462$ (−53.8%)

NORMALITY TEST · Validating Distribution Assumption for Cpk

Frequency Distribution of X-bar Values (Ω)



Bell-shaped, symmetric distribution → data is approximately normally distributed

ANDERSON-DARLING TEST

AD Statistic = 0.511

p-value = **0.178** > 0.05

INTERPRETATION

- Histogram shows bell-shaped distribution
- Data follows normal probability line closely
- p-value (0.178) > α (0.05) → Fail to reject H_0
- ✓ **Normality SATISFIED** → Cpk analysis is valid

PROCESS CAPABILITY ANALYSIS · Cpk

CALCULATIONS

LSL (Lower Spec Limit)	4.20 Ω
USL (Upper Spec Limit)	4.80 Ω
Final Process Mean ($\bar{X}$)	4.50346 Ω
Final Avg Range (R)	0.146 Ω
d_2 (for $n=5$, Appendix VI)	2.326
$\sigma = R / d_2 = 0.146 / 2.326$	0.0628
$C_{pu} = (USL - \bar{X}) / 3\sigma$	1.57
$C_{pl} = (\bar{X} - LSL) / 3\sigma$	1.61
$Cpk = \min(C_{pu}, C_{pl})$	1.57

Cpk RESULT

1.57

Process Capability Index (Cpk) · Manual Calculation

Cpk < 1.0

INCAPABLE

1.0 - 1.33

MARGINAL

> 1.33 ✓

Minitab Validation

Cpk (Minitab) $\approx$ **1.69** · Both exceed 1.33 → **HIGHLY CAPABLE**

PROCESS CAPABILITY ANALYSIS · Cpk

CALCULATIONS

LSL (Lower Spec Limit) 4.20 Ω

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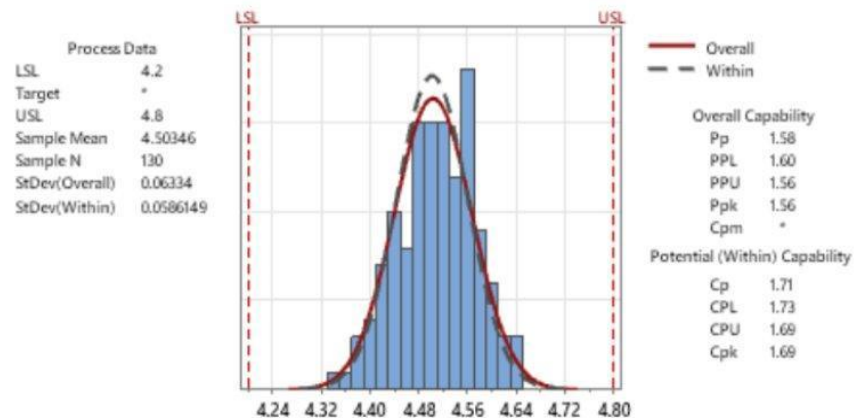
$C_{pu} = (USL - \bar{X}) / 3\sigma$ 1.57

$C_{pl} = (\bar{X} - LSL) / 3\sigma$ 1.61

$Cpk = \min(C_{pu}, C_{pl})$ 1.57

MINITAB PROCESS CAPABILITY REPORT

Process Capability Report for WR1 (Ω), ..., WR5 (Ω)



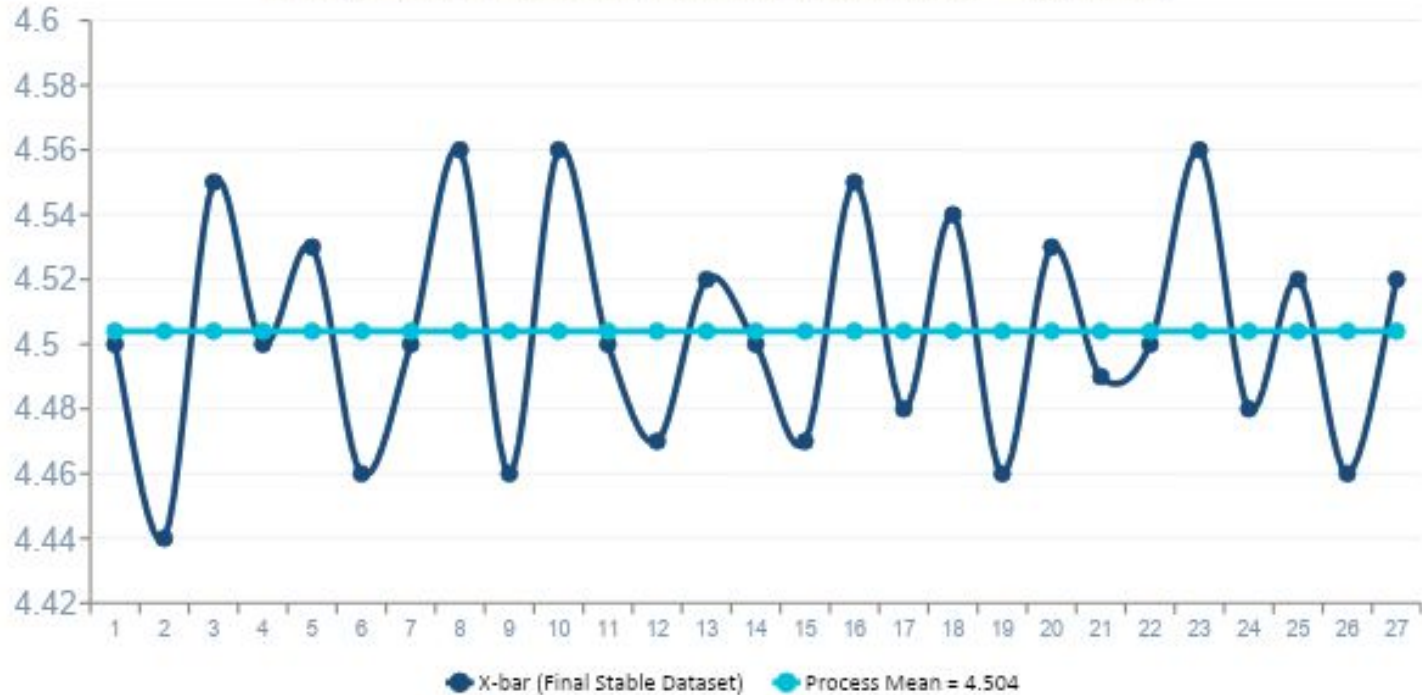
	Performance		
	Observed	Expected Overall	Expected Within
PPM < LSL	0.00	0.83	0.11
PPM > USL	0.00	1.42	0.21
PPM Total	0.00	2.25	0.32

The actual process spread is represented by 6 sigma.

✓ Cpk = 1.57 (Manual) · Cpk = 1.69 (Minitab) — Both exceed 1.33 → HIGHLY CAPABLE PROCESS

RUN CHART • Process Behavior Over Time

Run Chart: X-bar Values over Batches (Final Stable Dataset — 27 Batches)



FINDINGS

✓ No Trends

No systematic drift up or down

✓ No Shifts

No prolonged mean displacement

✓ Random Pattern

Points fluctuate around CL randomly

✓ Stable Process

Consistent with X-bar/R charts

PROCESS PERFORMANCE · Before vs After Improvement

METRIC	BEFORE IMPROVEMENT	AFTER IMPROVEMENT
R Chart	✗ B16 exceeded UCL ($R = 0.380 > 0.326$)	✓ All 29 points within control limits
X-bar Chart	✗ B13 (4.720) & B19 (4.650) above UCL	✓ All 27 points within control limits
P-Chart	✗ B14 & B19 exceeded UCL ($p = 0.60$)	✓ All points within revised limits
Average Defect Rate ($\bar{p}$)	10.0% ($\bar{p} = 0.10$)	4.62% ($\bar{p} = 0.0462$)
Process Capability (Cpk)	✗ Not reliable (process unstable)	✓ Cpk = 1.57 (manual) 1.69 (Minitab)
Statistical Control Status	✗ NOT in statistical control	✓ FULLY in statistical control

FINAL CONCLUSION

01

Process Was Initially Unstable

Control charts revealed out-of-control points (B13, B14, B16, B19). Both process mean and variation were affected by special causes.

02

Root Causes Identified

Fishbone & Pareto analysis: Thermal Issues (45.5%) and Material Defects (27.3%) account for 72.7% of all defects.

03

Corrective Actions Implemented

Thermal control, material quality standardization, preventive maintenance, and operator training were successfully applied.

04

Process Achieved Full Control

Revised charts confirmed stability. $Cpk = 1.57-1.69 (> 1.33)$. Defect rate dropped from 10% to 4.62%.

THANK YOU

We are happy to answer any questions.

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● Mohamad Sherif Shabrawy

● Mohamed Ali Ismail

● Verina Elkess

● Yousef Wail Ismail

● Amr Sherif Maher